



Shiyu Shen

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Research Profile

- My research focuses on uncertainty in machine learning, Bayesian deep learning, and domain generalization.
- In-depth understanding of the foundations of diffusion/flow models and LLMs.
- Proficient in PyTorch, verl, and slime, with hands-on experience in large-scale pretraining, SFT and RL.
- Extensive project experience spanning model development, system integration, and deployment.

Education

Nankai University	Bachelor's degree; Gongneng Scholarship	2017–2021
Nankai University	Master's degree; Gongneng Scholarship, Zhou Enlai Scholarship, National Scholarship	2021–2024
Nankai University	Ph.D. studies; National Scholarship	2024–2028

Publications & Preprints

1. Z. Wang, B. Pan, **Shiyu Shen**, T. Shi, and Z. Shi. "Domain Generalization Guided by Large-Scale Pre-Trained Priors." *arXiv preprint arXiv:2406.05628*, 2024.
2. Y. Tan, J. Wu, Y. Liu, **Shiyu Shen**, X. Xu, and B. Pan. "A Lightweight Transformer-Based Spatiotemporal Analysis Prediction Algorithm for High-Dimensional Meteorological Data." *Remote Sensing*, 2024.
3. **Shiyu Shen**, B. Pan, Y. Wang, Y. Liu, and Z. Shi. "A Second Order Stationarity Based Confidence Assessment Method for Temperature Forecast." *IEEE Geoscience and Remote Sensing Letters*, 2025.
4. **Shiyu Shen**, B. Pan, Z. Zhang, and Z. Shi. "Hyperspectral Image Generation with Unmixing Guided Diffusion Model." *arXiv preprint arXiv:2506.02601*, 2025. *Under review at IEEE TGRS*.
5. **Shiyu Shen**, B. Pan, and G. Xue. "A Pretrained Probabilistic Transformer for City-Scale Traffic Volume Prediction." *arXiv preprint arXiv:2506.02654*, 2025. *Under review at IEEE TMM*.
6. **Shiyu Shen**, Z. Gao, T. Chai, Y. Huang, and B. Pan. "Contrastive Heliophysical Image Pretraining for Solar Dynamics Observatory Records." *arXiv preprint arXiv:2511.22958*, 2025.
7. Z. Gao, **Shiyu Shen**, X. Xiang, W. Li, Y. Gao, and Q. Fan. "SmartSAM: Segment Ambiguous Objects like Smart Annotators." 2025.
8. **Shiyu Shen**, B. Pan, T. Shi, T. Li, and Z. Shi. "Be Bayesian by Attachments to Catch More Uncertainty." *Pattern Recognition*, 2026.
9. B. Pan, **Shiyu Shen**, Z. Wang, Z. Shi, and X. Xu. "Preserving Domain Generalization in Fine-Tuning via Joint Parameter Selection." *IEEE Transactions on Multimedia*, 2026.
10. Z. Gao, **Shiyu Shen**, T. Chai, W. Wang, H. Xu, W. Li, et al. "VideoTIR: Accurate Understanding for Long Videos with Efficient Tool-Integrated Reasoning." *European Conference on Computer Vision (ECCV)*, 2026.
11. **Shiyu Shen**, B. Pan, T. Shi, T. Li, and Z. Shi. "Bayesian Domain Invariant Learning via Posterior Generalization of Parameter Distributions." *arXiv preprint arXiv:2310.16277*, 2026. *Under review at Pattern Recognition*.
12. **Shiyu Shen**, B. Pan, Z. Wang, and Z. Shi. "A Distribution-Free Efficient Bayesian Framework for Domain Generalization." 2026. *Under review at IEEE TMM*.

Research & Engineering Projects

UAV Electro-Optical Imaging Teaching System:	Scene-generation subsystem.	2024
Maritime Optical Data Analysis System:	Confidence-assessment subsystem.	2024
Inertial Navigation High-Order Error Identification:	Project lead; big-data regression analysis.	2025
Virtual-Real Intelligent Traffic Simulation Platform:	Traffic-volume prediction and signal control.	2025

Research & Industry Experience

Tianrang — TRFold		2021
Yunqi Engineering Institute — Intelligent Transportation:	Train traffic prediction and signal-control models.	2024
Zhejiang Lab — AstrOne:	Lead SolarCLIP and SolarGPT development.	2024
Zhejiang Lab — 021 Chemical LLM:	Lead SFT and GRPO; design curriculum learning.	2025
Zhejiang Lab — 021 Mathematics LLM:	Build CPT data.	2025
Zhejiang Lab — 021 Code Agent:	Design stepwise verifiable RL rewards.	2026–Present